

AIRFARE PRICE PREDICTION

ABSTRACT

Accurate forecasting of flight ticket fares is essential for optimizing planning and revenue for travelers and airlines. Traditional models often fall short in capturing the complex temporal dependencies and external factors influencing ticket prices. This project proposes an enhanced prediction system using a hybrid deep learning model that combines Gated Recurrent Units (GRU) and Long Short-Term Memory (LSTM) networks. By integrating GRUs' efficient computation with LSTMs' superior memory capabilities, the hybrid model captures intricate patterns and dependencies more effectively. Enhanced feature engineering incorporates additional data, such as weather, events, economic indicators, and historical fare trends, while an attention mechanism improves interpretability and performance. Advanced hyperparameter tuning ensures optimal performance, and real-time forecasting capabilities provide up-to-date fare predictions. Using technologies like Python, TensorFlow, Keras, and Scikit-learn, this hybrid GRU-LSTM model aims to deliver a more accurate and robust flight fare forecasting system, offering valuable insights for consumers and airlines.